

Eyeris: Machine Learning Powered Assistive Glasses for the Visually Impaired

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Abstract

For the over 200 million individuals around the world with visual impairments, navigation remains a persistent challenge due to restricted sensory inputs, dependence on others, and safety risks. Existing technology such as white canes, or GPS-based navigation apps struggle with accurate scene comprehension, contextual awareness, and adaptability to dynamic settings. Recent advancements in embedded GPU acceleration, high-performance vision language models (VLMs), and large language models (LLMs) enable a huge lead in assistive device technologies. This project introduces Eyeris, a wearable assistive glasses system powered by the multimodal vision model Llama3.2-Vision, designed to empower visually impaired individuals through real-time environmental understanding and intuitive audio feedback.

Eyeris integrates a 360-degree camera for peripheral vision coverage and Llama3.2-Vision quantized to INT8 precision for low latency and offline personal assistant capabilities. The system processes visual data through a hybrid Vision Transformer (ViT) and transformer-based text encoder, generating contextualized descriptions of obstacles, signs, and hazards. A voice-activated interface allows natural queries with responses synthesized to speech via the gTTS text-to-speech engine.

By merging real-time 360-degree perception with LLM-driven reasoning, Eyeris advances independence, safety, and confidence for visually impaired users, marking a shift in assistive technology. This work highlights the potential of machine learning to reduce accessibility gaps.

Keywords

Visual Impairments, Assistive Technology, Quantization, Large Language Models, Vision Transformer

Ccs Concepts

Software and its engineering~Software design engineering;
Computer systems organization~Real-time system architecture;
Computing methodologies~Scene understanding

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1. Introduction

Deep learning has revolutionized the approach towards complex classification problems. Deep neural networks learn hierarchical representations of data enabling them to detect patterns and relationships that conventional machine learning techniques may miss. Over the past decade, deep learning has boosted performance in many fields such as healthcare (medical image diagnoses, personalized treatment plans), finance (risk analysis, fraud detection), and computer vision (facial recognition, object detection, and scene understanding) [1].

The ability to classify images accurately and efficiently becomes even more significant in the context of assistive technologies. Visually impaired individuals need reliable identification of everyday objects, navigation cues, and environmental context to move about safely and confidently. By integrating deep neural network technology into wearable systems, such as glasses, real-time feedback can mitigate many of the daily challenges faced by the visually impaired and thus lead to more independence and improved quality of life [2].

The proposed idea is to research and develop assistive glasses for the visually impaired individuals, utilizing AI and deep learning methods for real time scene understanding. The algorithm will structure data from the surroundings which would then be fed through a Text-To-Speech algorithm to communicate with audio cues. The primary motivation behind creating assistive glasses is to enhance the independence, quality of life, and overall well-being of visually impaired individuals. By increasing their autonomy in navigation and information processing, the product would have significant social benefits.

2. Literature Review

Machine learning is a field of artificial intelligence that focuses on the learning aspects to develop algorithms that best represent a set of data. Instead of a computer being provided with a dataset and an algorithm, the computer is provided with a dataset and a few example outputs. The computer then recognizes patterns to develop its own algorithms. Deep learning networks are computational structures inspired by the human brain. They consist of layers of interconnected “neurons” each performing a weighted sum of its input followed by an activation function to produce an output. The weights are iteratively adjusted through back propagation to minimize a loss function allowing the network to recognize and learn complex patterns [1].

Convolution Neural Networks (CNNs) have become the dominant architecture for image classification due to its high performance and efficiency. CNNs use filters called kernels that slide over small dimensions of the input. Each kernel is trained to extract a specific type of feature (texture, shape, color) from the regions of the input. By stacking multiple convolution layers, each having a broader scope, the neural network is able to accurately identify more complex and abstract features [2].

Computer vision systems are one of the earliest and most well recognized machine learning technologies. Recent developments in computer vision, such as attention mechanisms, aim to imitate the natural aspects of the human vision system [3]. This led to the application of computer vision systems in a variety of social, economic, and industrial solutions. A few popular examples of these systems are road scene understanding, medical imaging analysis, and assistive technologies [4].

Machine learning has emerged as a powerful tool for advancing assistive technologies, particularly for individuals with visual impairments [5]. Recent studies indicate that ML-driven algorithms can accurately recognize objects, detect obstacles, and transform visual data into speech or tactile feedback, thereby enhancing user autonomy and safety. Such capabilities underscore the promise of ML in making daily navigation and environmental awareness more accessible for blind or low-vision users.

Over the past decade, machine learning has evolved from more statistical use cases to powerful deep learning models capable of handling vast amounts of unstructured data. Modern Large Language Models (LLMs) use transformer architectures to grasp nuanced language architectures, enabling them to formulate long coherent text in almost real time [6].

LLMs have transitioned from text centric systems to multimodal architectures capable of processing and integrating diverse data types, including images, video, and audio. This evolution has been driven by the need for AI systems to understand and generate content across multiple modalities, thereby enhancing their applicability in real world scenarios. Early advancements in vision-language integration involved models that combined visual and textual data to perform tasks such as image captioning and visual question answering. These models laid the groundwork for more sophisticated systems that could handle complex multimodal interactions. Specifically, the development of Vision Transformers (ViT) has significantly improved performance over previous vision language models by leveraging large scale pretraining and transfer learning techniques [7].

Shen et al. developed a wearable assistive device that combines real-time object detection with haptic feedback to improve navigation capabilities for blind individuals. They utilized computer vision algorithms to classify objects targeted for pedestrians (vehicles, street furniture, pedestrians) and communicate the information to the user through haptic feedback through gloves or wristbands. The user testing demonstrated increased environmental awareness and improved obstacle avoidance [8].

Neto et al. developed a kinetic-based face recognition system aimed at assisting visually impaired users in identifying individuals in their vicinity. They used Microsoft Kinetic sensors and face recognition algorithms to process facial data in real time. The paper emphasized the system's accuracy and responsiveness highlighting its proficiency in social interactions and personal safety [9].

Mann et al. developed a wearable navigation aid that integrates a range camera with a vibrotactile helmet to enhance obstacle detection for blind individuals. The camera captures environmental data, which is translated into tactile signals

communicated through the helmet. The trials indicated effective obstacle avoidance and improved navigation capabilities [10].

The proposed technology distinguishes itself by incorporating LLMs to provide real-time conversational outputs, enabling dynamic assistance. Unlike previous studies relying solely on tactile feedback, this approach allows for interactive dialogue, allowing users to review detailed descriptions, and obtain personalized information about their environment.

3. Methodology

The system follows a three-stage pipeline. Stage 1: A mounted camera captures real time RGB frames which are encoded as base64 strings for compatibility with Llama3.2-Vision. In parallel, user queries are transcribed to text through a speech recognition model. Stage 2: Both the query and image are then fed to the LLM along with a prewritten prompt coaching the LLM on its task. The LLM is quantized to INT4 precision, reducing the computational power and allowing it to run on less powerful GPUs. Stage 3: The LLMs output is synthesized to speech through a text-to-speech engine and relayed to the user through headphones.

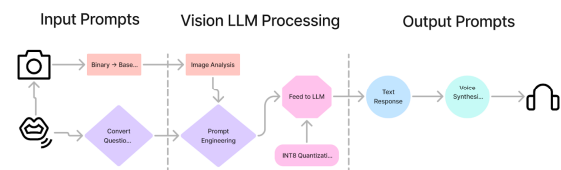


Figure 1: Proposed Pipeline for Eyeris

3.1 Input Prompts

The device requires a voice command from the user to trigger the algorithm. The voice command is converted to a text format in order to be conveyed to the LLM.

3.1.1 Audio to Text Prompts

The device uses a 2-layer speech recognition algorithm using Google text to speech. The primary layer listens for the voice command “Iris”. When triggered, the secondary layer then listens for 10 seconds while the user asks a short question and a prompt regarding the surroundings. 10 seconds has been studied to be the average time to provide one liner.

3.1.2 Visual Prompts

During the model inference, we use real time indoor and outdoor scenes. Also triggered by the command word, a mounted camera captures a 256x256px RGB image of the surrounding scene and converts it to a base64 input embedding for Llama3.2-Vision.

3.2 Data preprocessing and Augmentation

For image Preprocessing, we have experimented with different dimensionality of input images. Higher image resolutions do offer better feature selection and object visibility, but it also leads to heavy computational costs that eventually lead to slower inference times. In real time navigation systems, slower inference times lead to harmful outcomes. Therefore, we settled on 128x128. This size offers enough input features for the LLM to comprehend the scene and offer a detailed insight.

3.3 Vision LLM Processing

The system uses Llama3.2-Vision, 11B parameter model for its unified processing of visual and textual data. The model integrates a Vision Transformer (ViT) encoder, which partitions input images into 16x16 patches, generating 384-dimensional embeddings. The embeddings are fused with text tokens through 12 cross-attention layers, enabling reasoning over visual and linguistic inputs. Comparative benchmarks show that Llama3.2-Vision outperforms smaller models such as Llava by 14.2% in low light tasks, justifying its suitability for assistive applications.

In order to achieve real-time classification, the model is quantized to INT8 precision using Unsloth AI, reducing its memory down to 5 GB while retaining 98.5% accuracy. The optimization enables sub-500ms inference latency on devices such as the NVIDIA Jetson Nano.

Unsloth AI is a web-based platform that specializes in fine-tuning large-scale language models to improve computational costs and computing time. Unsloth AI uniquely offers 4bit quantization, much faster than traditional quantization, leading to a significant reduction in model size while maintaining minimal loss in accuracy.

By leveraging Unsloth's quantized models, we can design and deploy sophisticated visual guidance and explanation systems for blind users. Unsloth AI offers efficiency and high accuracy fine-tuning accurately, enhancing accessibility. Moreover, the reduced model size facilitates integration of these systems into devices with limited computational resources ensuring that visually impaired users can benefit from real time descriptive feedback without requiring high end hardware.

3.4 Output Prompts

Llama3.2's output is a text response covering all the necessary details about the image capture to the user. Text to Audio Conversion

In order to relay the text to the user in an audio format, the algorithm uses Google Text-to-Speech (gTTS). The model is able to analyze the text and output an mp3 file with sub 3sec latency. The mp3 file is then relayed to the user via speakers.

4. Experimental Results

4.1 Experimental Setup

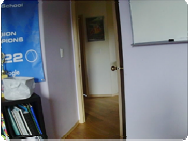
The programmed pipeline uses PyTorch as the machine learning library. The program was run on an NVIDIA 4060 GPU and AMD Ryzen 5 CPU. We used a standard mix and speakers for audio inputs and outputs.

4.2 Performance Evaluation and Analysis

The pipeline receives an input image along with an audio prompt that is later converted to a textual embedding. Both embeddings are then fed to the finetuned Vision LLM that produces a textual description of the scene which can be translated to audio signals by the voice synthesizer. To evaluate the performance of the output textual prompts, we have done manual assessments of the prompts against the input images. The multiple tests have shown a very high accuracy of scene description. A few examples are shown in Table 1 below.

Table 1: Example Inputs, Queries, and Outputs from Eyeris

Input Image	Query	Output
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	How can I leave my room?	"The door in front of you is slightly open. The door handle is located slightly to the right. As you approach there will be a wall on the left as well as a shelf with a few objects on it."
	What does the scenery look like	"Out your window, there is a gray basketball hoop next to a blue car parked in the driveway. The surroundings are residential neighborhoods with grassy lawns and bushes. When stepping outside, you may want more assistance crossing the road. Feel free to ask me again for more assistance."
	Is it safe to cross the road?	"You are currently at a crosswalk. Walking to the other side of the road is a singular pedestrian in black clothes and a blue backpack. There are currently no cars approaching the intersection, so it is safe to proceed."
	What's in front of me?	"The wall is covered in graffiti with the text 'The Land Fresh' off to the right. The rest of the graffiti is hard to interpret but might depict a bird symbolizing freedom."

4.3 Discussion of Findings

Table 4.1 shows the input image, the User query, and the possible output response. Eyeris was given multiple indoor and outdoor real-life scenes and based on short, easy prompts, the LLM generates a very precise and informative output which could be useful for the visually impaired population. The key idea for this pipeline is to keep the input query and images as simple as possible to generate a very short and detailed description of the scene that would help visually disabled people navigate with more confidence and freedom. Our results show that the proposed pipeline is effectively able to achieve our goals. The pipeline has been optimized to offer minimal inference times. The response time for each query processing was around 1-3 seconds, which is incredibly effective for real time navigation.

5. Conclusion

The proposed research has offered a unique idea to improve the quality of life for visually impaired people. By taking advantage

of recent developments in large scale language models that have transformed every aspect of our lives, we proposed a pipeline that helps the visually impaired navigate in real settings. The proposed pipeline uses vision LLMs and produces very fast inferred output prompts that can be converted to audio signals to help navigate the visually impaired. The architecture is designed and developed on traditional CPU and GPU settings of a personal computer. Our future research is relevant to designing a portable system including NVIDIA Jetson as well as mobile apps. Our key focus is to make this system available to the wider public by hosting the framed vision LLM on cloud servers which will eventually improve the inference and access of the device.

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